(Assignment No 1)

Mth632

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Student ID:

Question#1:

Express the complex number $(-\sqrt{3}+i)^2$ in exponential form.

To express the complex number $(-\sqrt{3}+i)^2$ in exponential form, we first find the polar form of $(-\sqrt{3}+i)$ and then square it.

1. Find the modulus (r):

$$r = |z| = \sqrt{(-\sqrt{3})^2 + 1^2} = \sqrt{3+1} = \sqrt{4} = 2$$

2. Find the argument θ:

We use the fact that the complex number is in the second quadrant (since $(-\sqrt{3})$ is negative and (i) is positive). The argument (θ) is given by:

$$\theta = \arg(z) = \pi - \tan^{-1}\left(\frac{1}{\sqrt{3}}\right) = \pi - \frac{\pi}{6} = \frac{5\pi}{6}$$

Thus, we can write:

$$z = -\sqrt{3} + i = 2\left(\cos\left(\frac{5\pi}{6}\right) + i\sin\left(\frac{5\pi}{6}\right)\right)$$

3. Square the complex number:

Now square (z):

$$z^2 = \left(2\left(\cos\left(\frac{5\pi}{6}\right) + i\sin\left(\frac{5\pi}{6}\right)\right)\right)^2$$

Using the formula for powers of complex numbers in polar form:

$$z^2 = 2^2\left(\cos\left(2\cdot\frac{5\pi}{6}\right)+i\sin\left(2\cdot\frac{5\pi}{6}\right)\right)$$

$$z^2 = 4\left(\cos\left(\frac{5\pi}{3}\right)+i\sin\left(\frac{5\pi}{3}\right)\right)$$

Thus, the exponential form of (z^2) is:

$$z^2 = 4e^{i\frac{5\pi}{3}}$$

Question No 2:

Solution:

To express $(f(x)=\overline{z}\,\mathrm{Im}(z)+z\,\mathrm{Re}(z)+z\overline{z})$ in the form $(f(z) = u(x, y) + iv(x, y))$, where $(z=x+iy)$ *and* $(\overline{z}=x-iy)$, let's analyze each term and simplify.

1. Write (z) and $(\overline{z})$ in terms of (x) and (y):

$(z=x+iy)$
$(\overline{z}=x-iy)$

The imaginary part of (z), $(\mathrm{Im}(z)=y)$

The real part of (z), $(\mathrm{Re}(z)=x)$

2. Substitute and simplify each term:

First term: $(\overline{z}\,\mathrm{Im}(z))$

$$\overline{z}\,\mathrm{Im}(z)=(x-iy)y=xy-iy^2$$

Second term: $(z\,\mathrm{Re}(z))$

$$z\,\mathrm{Re}(z)=(x+iy)x=x^2+ixy$$

Third term: $(z\overline{z})$

$$z\overline{z}=(x+iy)(x-iy)=x^2+y^2$$

3. Combine all terms to form (f(z)):

Now, we add up the three expressions we obtained:

$$f(z)=\overline{z}\,\mathrm{Im}(z)+z\,\mathrm{Re}(z)+z\overline{z}$$

$$f(z) = (xy - iy^2) + (x^2 + ixy) + (x^2 + y^2)$$

Group real and imaginary parts:

Real part (u(x, y)):

$$u(x, y) = xy + x^2 + x^2 + y^2 = 2x^2 + xy + y^2$$

Imaginary part (v(x, y)):

$$v(x, y) = -y^2 + xy = xy - y^2$$

Q3: Find 4th roots of -1.
Solution:

Step 1: Express (-1) in Polar Form

The complex number (-1) has a modulus of (1) (since (|-1| = 1)) and an argument of (π) (since it lies on the negative real axis).

So, we can write:

$-1 = 1 \cdot \left(\cos(\pi) + i\sin(\pi)\right)$ or in exponential form,

$$-1 = e^{i\pi}$$

Step 2: Apply De Moivre's Theorem for the 4th Roots

The fourth roots of a complex number $(z = re^{i\theta})$ are given by:

$$w_k = r^{1/4} e^{i\left(\frac{\theta + 2k\pi}{4}\right)} \quad \text{for} \quad k = 0, 1, 2, 3$$

In our case, (r = 1) and $(\theta = \pi)$, so the 4th roots are:

$$w_k = 1^{1/4} e^{i\left(\frac{\pi + 2k\pi}{4}\right)} = e^{i\frac{\pi + 2k\pi}{4}} \quad \text{for} \quad k = 0, 1, 2, 3$$

Step 3: Calculate Each Root

Now, we calculate each root by substituting (k = 0, 1, 2, 3):

1. For (k = 0):

$$w_0 = e^{i\frac{\pi}{4}} = \cos\left(\frac{\pi}{4}\right) + i\sin\left(\frac{\pi}{4}\right) = \frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2}$$

2. For (k = 1):

$$w_1 = e^{i\frac{3\pi}{4}} = \cos\left(\frac{3\pi}{4}\right) + i\sin\left(\frac{3\pi}{4}\right) = -\frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2}$$

3. For (k = 2):

$$w_2 = e^{i\frac{5\pi}{4}} = \cos\left(\frac{5\pi}{4}\right) + i\sin\left(\frac{5\pi}{4}\right) = -\frac{\sqrt{2}}{2} - i\frac{\sqrt{2}}{2}$$

4. For (k = 3):

$$w_3 = e^{i\frac{7\pi}{4}} = \cos\left(\frac{7\pi}{4}\right) + i\sin\left(\frac{7\pi}{4}\right) = \frac{\sqrt{2}}{2} - i\frac{\sqrt{2}}{2}$$

Answer

The four fourth roots of (-1) are:

$$w_0 = \frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2}, \quad w_1 = -\frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2}, \quad w_2 = -\frac{\sqrt{2}}{2} - i\frac{\sqrt{2}}{2}, \quad w_3 = \frac{\sqrt{2}}{2} - i\frac{\sqrt{2}}{2}$$

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Regards

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